

A CERTIFIED CONTACT-GEOMETRIC LOWER BOUND FOR THE THREE-DIMENSIONAL BLASCHKE–LEBESGUE PROBLEM

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ABSTRACT. Let $K \subset \mathbb{R}^3$ be a convex body of constant width $d > 0$. We give an exact-rational computer-assisted certificate for the universal estimate

$$\text{Vol}(K) \geq \frac{26220940089713\pi}{20000000000000} d^3 = 0.4118775637803022742481789259828398710025 \dots d^3.$$

The proof divides the possible circumradii at the rational value $R_* = \frac{305695601}{500000000}$. Below R_* , a finite radial-propagation certificate refines the contact-geometric method of HYRA. Above R_* , four balanced circumsphere contacts force a high-resolution polyhedral core; a sharp affine determinant transfer, a dominant-edge coordinate-width estimate, and six pairwise disjoint cone caps complete the argument. Every claim-bearing finite comparison is checked with integers and rational arithmetic. A binary64 Python implementation and a base-R implementation are independent numerical audits, and an adversarial suite verifies rejection of seven classes of corrupted proof objects. This draft does not address equality, sharpness, the identity of a minimizer, or the Meissner conjecture.

Verification status. This is an unreleased computer-assisted research draft. The standalone exact verifier, independent Python and base-R audits, reconstruction tests, and adversarial mutation tests have passed in the accompanying package. A public release, final authorship metadata, external mathematical review, and peer review are not asserted. The exact certificate JSON and exact verifier are the proof authority for the finite computations.

1. INTRODUCTION

A convex body has constant width d when every pair of parallel supporting planes orthogonal to a unit direction is separated by the same distance d . In the plane, the Blaschke–Lebesgue theorem identifies the Reuleaux triangle as the least-area body of prescribed constant width [9, 1]. The analogous three-dimensional problem is open. The two classical Meissner bodies have common volume

$$V_M(d) = \frac{\pi}{12} \left(8 - 3\sqrt{3} \arccos \frac{1}{3} \right) d^3 = 0.419860045965080 \dots d^3,$$

and are conjectured to minimize volume; this explicit comparison value is not a proved universal minimum [7, 10].

Chakerian proved the lower bound $\frac{\pi}{3}(3\sqrt{6} - 7)d^3 = 0.364916122595 \dots d^3$ [2]. Nishioka later obtained $\frac{4\pi}{33}d^3 = 0.380799109526 \dots d^3$ from a support-function formulation, spherical harmonics, and Bochner’s formula [11]. A subsequent contact-geometric manuscript by HYRA combined the minimum circumsphere, radial propagation, a divergence estimate, and exact cap recovery to reach a certified coefficient above 0.411040473721188 [5]. The present certificate develops that contact-geometric route further.

The main claim of the package is the following.

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Theorem 1.1 (Main computer-assisted claim). *Every convex body $K \subset \mathbb{R}^3$ of constant width $d > 0$ satisfies*

$$\text{Vol}(K) \geq \frac{26220940089713\pi}{20000000000000} d^3 = 0.4118775637803022742481789259828398710025 \dots d^3.$$

The argument is homogeneous, so we work at width one. If R is the radius of a minimum circumscribed ball, Jung's theorem gives

$$\frac{1}{2} \leq R \leq \frac{\sqrt{6}}{4}.$$

The certificate uses the exact handoff

$$R_* = \frac{305695601}{500000000} = 0.611391202.$$

A finite radial certificate covers $R \leq R_*$. The complementary interval is handled by a nearly tetrahedral four-contact configuration, a 98,306-ray forced fan, and six additional cone caps. The near-Jung branch gives the strictly larger lower value $0.4118796712148324317327979810 \dots$, leaving the final row of the radial partition as the unique bottleneck.

The proof has two layers. The geometric reductions are stated and proved in the text. The finite claims are represented by immutable integer and rational objects and replayed by the exact verifier. Section 9 records the trust boundary and the division between proof authority and independent audits.

2. CONSTANT-WIDTH IDENTITIES AND MINIMUM CIRCUMSPHERE CONTACTS

Let h_K be the support function of a width-one body K . Then

$$h_K(u) + h_K(-u) = 1 \quad (u \in \mathbb{S}^2). \quad (1)$$

Consequently

$$K + (-K) = B(0, 1), \quad \text{diam } K = 1. \quad (2)$$

In dimension three, Blaschke's identity relates volume and surface area:

$$S(K) = 2 \text{Vol}(K) + \frac{2\pi}{3}. \quad (3)$$

These standard identities are reviewed in [3, 10, 12].

Translate a minimum circumscribed ball to $B(0, R)$ and let $F = K \cap \partial B(0, R)$. Minimality of the ball implies $0 \in \text{conv } F$. Caratheodory's theorem therefore gives contact directions $u_i \in \mathbb{S}^2$ and positive weights λ_i such that

$$p_i = Ru_i \in K, \quad \sum_i \lambda_i = 1, \quad \sum_i \lambda_i p_i = 0, \quad (4)$$

using at most four contacts. Since $\text{diam } K = 1$, $|p_i - p_j| \leq 1$. The usual balance argument yields $R^2 \leq 3/8$. When $R^2 > 1/3$, four contacts are necessary; this is the regime used in the near-Jung branch.

Constant width also supplies an opposite point at each support point.

Lemma 2.1 (Opposite-point identity). *If $x \in \partial K$ has an outer unit normal n , then $x - n \in K$. Hence, when $K \subset B(0, R)$,*

$$\langle x, n \rangle \geq \frac{|x|^2 + 1 - R^2}{2}. \quad (5)$$

Proof. Let y lie on the supporting plane opposite x in direction n . Width one gives $\langle x - y, n \rangle = 1$, whereas (2) gives $|x - y| \leq 1$. Equality in Cauchy–Schwarz forces $x - y = n$. Since $x - n \in K \subset B(0, R)$, expanding $|x - n|^2 \leq R^2$ gives (5). $\square$

Because $R < 1$, the circumcenter lies in the interior of K . We write $\rho : \mathbb{S}^2 \rightarrow (0, \infty)$ for the radial function about that center.

3. THE LOWER-CIRCUMRADIUS RADIAL CERTIFICATE

This section records the analytic reduction and the finite exact data used on $[1/2, R_*]$. The construction follows the contact-geometric framework in [5], with a refined terminal partition.

3.1. Transport inequality and tangent gap. For $0 < m \leq 1 - R^2$, define

$$F_m(x) = \frac{2x}{|x|^2 + m}.$$

By Lemma 2.1, $\langle F_m(x), n \rangle \geq 1$ on every smooth support point. Applying the divergence theorem and then spherical integration gives

$$3V = \int_{\mathbb{S}^2} \rho^3 d\sigma, \quad S \leq \int_{\mathbb{S}^2} \frac{2\rho^3}{\rho^2 + m} d\sigma, \quad (6)$$

first for C^1 bodies and then for arbitrary constant-width bodies by outer parallel regularization.

Put

$$\varphi_m(X) = \frac{2X}{X^{2/3} + m} \quad (X > 0).$$

A direct differentiation shows $\varphi_m'' < 0$. If $aX + b$ is the tangent at $X = s_0^3$, the gap

$$G_{m,s_0}(s) = as^3 + b - \frac{2s^3}{s^2 + m}$$

is nonnegative, vanishes only at s_0 , and is monotone away from s_0 . Writing $E_G = \int_{\mathbb{S}^2} G_{m,s_0}(\rho) d\sigma$, equations (3) and (6) imply

$$V \geq \frac{2\pi/3 - 4\pi b + E_G}{3a - 2}, \quad 3a - 2 > 0. \quad (7)$$

Thus the purely analytic bound discards $E_G \geq 0$, while a stronger finite certificate recovers part of the gap on forced spherical caps.

3.2. Rational radial propagation. Suppose $Ru_0 \in K$. Diameter one implies the antipodal ball estimate

$$\rho(v) \leq -R_*c + \sqrt{1 - R_*^2(1 - c^2)}, \quad c = -\langle v, u_0 \rangle \in [0, 1], \quad (8)$$

whenever $R_* \leq R$. In the other direction, the supporting-plane bound (5) propagates a radial lower estimate from one direction to a nearby direction. With

$$\cos \gamma_m(s) = \frac{s^2 + m}{2s},$$

the one-step upper propagation map is

$$N_m(s, \delta) = \frac{s(s^2 + m)}{(s^2 + m) \cos \delta - \sqrt{4s^2 - (s^2 + m)^2} \sin \delta}. \quad (9)$$

The certificate uses the contrapositive of this estimate to construct radial lower chains away from each contact.

Every angle is represented by a rational tangent-half-angle grid:

$$t_k = \frac{k}{N}, \quad \cos \theta_k = \frac{1 - t_k^2}{1 + t_k^2}, \quad \sin \theta_k = \frac{2t_k}{1 + t_k^2}. \quad (10)$$

After clearing the one square root in (9), each propagation step is reduced to signed integer polynomial inequalities. The same is true for (8), cap disjointness, and the tangent-gap sum. The resulting arrays therefore contain no floating-point proof step.

3.3. Finite partition. The exact lower branch contains twelve rows. Four doubled-angular-grid cells replace a formerly insufficient parent row; their angular denominator is 76,800 and their radial denominator is $4 \cdot 10^{10}$. The weakest replacement margin over the terminal coefficient of π is $1.8799274348 \times 10^{-5}$. Appendix A gives the complete gap-free table generated directly from the certificate JSON. Its unique weakest row is 03B02, ending exactly at R_* .

4. NEAR-JUNG CONTACT GEOMETRY

Assume now $R \geq R_*$ and use four contacts in (4). Put $q = R^2$ and define the six edge deficiencies

$$e_{ij} = 1 - |p_i - p_j|^2 \geq 0, \quad S = \sum_{i < j} e_{ij}. \quad (11)$$

The Gram-kernel relation is equivalent to

$$\lambda_i + \sum_{j \neq i} \lambda_j e_{ij} = 1 - 2q. \quad (12)$$

Summing over i gives

$$\sum_{i < j} (\lambda_i + \lambda_j) e_{ij} = 3 - 8q. \quad (13)$$

Equation (12) implies $\lambda_i \leq 1 - 2q$; hence $\lambda_i + \lambda_j \geq 4q - 1$. Therefore

$$\boxed{S \leq E(R) := \frac{3 - 8R^2}{4R^2 - 1}.} \quad (14)$$

The function $E(R)$ decreases on the Jung interval, so throughout the near-Jung branch

$$S \leq E_* := E(R_*) = 0.01939912533733453686615863855265 \dots$$

4.1. Affine reference coordinates. Let

$$v_0 = (1, 1, 1), \quad v_1 = (1, -1, -1), \quad v_2 = (-1, 1, -1), \quad v_3 = (-1, -1, 1).$$

There are an invertible linear map L and a balance-center vector c such that

$$p_i = L(v_i - c).$$

Set $A = 2\sqrt{2}L$ and $H = A^T A = 8L^T L$. The regular unit-edge tetrahedron corresponds to $H = I$. Solving the six edge equations gives

$$H_{00} = 1 + \frac{1}{2}(e_{01} - e_{02} - e_{03} - e_{12} - e_{13} + e_{23}), \quad (15)$$

$$H_{11} = 1 + \frac{1}{2}(-e_{01} + e_{02} - e_{03} - e_{12} + e_{13} - e_{23}), \quad (16)$$

$$H_{22} = 1 + \frac{1}{2}(-e_{01} - e_{02} + e_{03} + e_{12} - e_{13} - e_{23}). \quad (17)$$

The affine volume loss is controlled by a sharp scalar determinant estimate.

Proposition 4.1 (Determinant transfer). *For nonnegative deficiencies with $S < 3/2$,*

$$\det H \geq 1 - \frac{S}{2} - \frac{S^2}{2}. \quad (18)$$

Consequently, on the near-Jung branch,

$$\det H \geq 0.99011227429940592416253966614320 \dots$$

Proof. Use three edge vectors from one contact and write their normalized Gram matrix in terms of the six squared edge lengths $1 - e_{ij}$. Twice its determinant is $\det H$. Expanding exactly and subtracting the right side of (18) produces fifteen quadratic cross terms, each with coefficient at least one, together with twelve negative squarefree cubic terms, each with coefficient $-1/2$, and additional nonnegative cubic terms. Each pair of deficiency variables occurs in at most four of the negative triples.

For a negative triple abc ,

$$abc \leq \frac{S}{3}(ab + ac + bc).$$

The total negative cubic part is therefore at least $-(2S/3) \sum_{r < s} e_r e_s$, whereas the positive quadratic part is at least $\sum_{r < s} e_r e_s$. Since $S < 3/2$, the difference is nonnegative. The exact verifier reconstructs the polynomial and checks the complete coefficient and pair-incidence census rather than trusting a stored expansion. $\square$

5. THE HIGH-RESOLUTION FORCED FAN

The near-Jung core is represented in the regular reference coordinates. The unit sphere is subdivided by the cubical grid of resolution $N = 128$. After identifying shared face boundaries, the triangulation has

$$98,306 \text{ rays,} \quad 196,608 \text{ triangular cones,} \quad 294,912 \text{ edges.}$$

The order-four stabilizer of the distinguished contact edge has 24,833 ray orbits.

For every ray, the certificate stores an integer radial numerator and a dyadic S-procedure multiplier. The exact verifier reconstructs the grid, its triangulation, and the stabilizer orbits from first principles. It then checks every ray inequality over the full nonnegative deficiency simplex

$$e_{01} \geq e_{ij}, \quad \sum e_{ij} \leq E_*.$$

The raywise optimization is exact: its candidates are the vertices forced by the dominant-edge budget, and all remaining arithmetic is rational. Relative to the immutable reference fan, 18,229 radial numerators move inward while all inherited multiplier numerators remain fixed. The least nontrivial exact margin is

$$0.0000000002888904207092857876733697822441 \dots > 10^{-12}.$$

Let $Q_i \in \mathbb{Z}^3$ be the reconstructed cubical rays and let r_i be the certified rational radii. Summing the signed determinants $\det(Q_i, Q_j, Q_k) r_i r_j r_k / 6$ over the fixed cone triangulation gives the reference coefficient

$$C_{\text{fan}} = 0.5851824745606993360954839072 \dots$$

The affine determinant factor from Proposition 4.1 then gives the physical forced-core estimate

$$\boxed{V_{\text{core}} \geq 0.4117357103127771735489695298 \dots} \quad (19)$$

The stored volume numerator is independently reconstructed from all 196,608 cones.

6. DOMINANT-EDGE COORDINATE WIDTHS

Relabel the contacts so that e_{01} is a largest deficiency. The fan atlas includes ties, so this does not discard any configuration.

Proposition 6.1 (Coordinate-specific transformed widths). *If $e_{01} \geq e_{ij}$ for every edge, then*

$$H_{00} \leq 1 + \frac{S}{2}, \quad H_{11}, H_{22} \leq 1 + \frac{S}{6}.$$

If $g_j = \sqrt{(H^{-1})_{jj}}$, then

$$g_0 \geq \frac{1}{\sqrt{1+S/2}}, \quad g_1, g_2 \geq \frac{1}{\sqrt{1+S/6}}. \quad (20)$$

Proof. The first diagonal estimate follows immediately from (15). For the second,

$$S - 3[2(H_{11} - 1)] = 2(e_{01} - e_{02}) + 2(e_{01} - e_{13}) + 4(e_{03} + e_{12} + e_{23}) \geq 0.$$

The identity for H_{22} is obtained by interchanging the corresponding indices. Proposition 4.1 makes H positive definite, and Cauchy–Schwarz in the H -inner product gives $1 \leq H_{jj}(H^{-1})_{jj}$. $\square$

Using $S \leq E_*$ and exact rational lower enclosures for the square roots, the certificate obtains

$$(g_0, g_1, g_2) \geq (0.995185216461759158\dots, 0.998387315752931109\dots, 0.998387315752931109\dots).$$

The distinction between the dominant direction and the two transverse directions is essential for the cap contribution.

7. SIX OPTIMIZED CONE CAPS

For coordinate axis j , let $h_{j,+}$ and $h_{j,-}$ be the exact signed supports of the reference fan. Intersecting the fan with fixed rational planes below these supports gives exact convex section polygons of areas $A_{j,+}$ and $A_{j,-}$. If $t_{j,+}$ and $t_{j,-}$ are the remaining exterior support extensions, Proposition 6.1 gives

$$t_{j,+} + t_{j,-} \geq T_j := g_j - \frac{h_{j,+} + h_{j,-}}{2\sqrt{2}}. \quad (21)$$

A section of physical area A , lying a depth d below the fan support, and an exterior support extension t force a cone of volume at least

$$\phi_{A,d}(t) = \frac{At^3}{24(d+t)^2}. \quad (22)$$

For $t > 0$,

$$\phi''_{A,d}(t) = \frac{Ad^2t}{4(d+t)^4} > 0.$$

Thus, subject to equality in (21), each antipodal pair has a unique minimizing allocation. The exact verifier brackets each stationary point by rationals at distance at most 10^{-20} and proves opposite derivative signs at the two bracket endpoints.

TABLE 1. Certified physical cone-cap contributions at the handoff radius.

Axis	Width lower	Section vertices $+/-$	Pair contribution
0	0.995185216461759158...	110/84	0.000050275601362298174...
1	0.998387315752931109...	102/76	0.000046842650346480004...
2	0.998387315752931109...	102/76	0.000046842650346480004...
Total			0.0001439609020552581838284512...

The two caps associated with one axis lie in opposite support halfspaces. For distinct axes, the balance-center uncertainty and the matrix inequality $H \succeq (1 - E_*)I$ show that an overlap would force two signed coordinates to exceed the least fan support while every transformed body point remains in a fixed ellipsoid. The exact contradiction gap is

$$0.6378776497509377628400649501\dots > 0.$$

Hence all six cap interiors are pairwise disjoint; possible boundary contacts have zero volume.

8. UNIVERSAL ASSEMBLY

Adding (19) and the three disjoint antipodal cap-pair contributions gives

$$V_{\text{near}} \geq 0.4118796712148324317327979810 \dots$$

The exact margin over the upper rational enclosure of the coefficient in Theorem 1.1 is

$$0.0000021074345301574846190550 \dots > 0.$$

Therefore the near-Jung construction covers $[R_*, \sqrt{6}/4]$.

For $R \leq R_*$, the exact partition in Section 3 applies. Its unique weakest row is 03B02, whose exact coefficient is $\frac{26220940089713}{200000000000000}\pi$. The two branch endpoints agree exactly, so the union is the full Jung interval. Finally, dilation from width one to width d multiplies volume by d^3 , completing the computer-assisted proof of Theorem 1.1 under the trust boundary below.

9. CERTIFICATE ARCHITECTURE AND TRUST BOUNDARY

The public-format proof package is standalone: no path points into the private research repository, and the complete replay begins from the files committed with this manuscript.

Exact proof authority. The program `certificate/certify_universal_bound.py` reconstructs the claim using Python integers and `Fraction` arithmetic. NumPy is used to read immutable integer arrays, not to decide theorem inequalities. The verifier checks the lower radial rows, reconstructs the fan and all cone volumes, verifies the determinant census and coordinate-width identities, rebuilds the six section polygons, proves the cap-allocation brackets and six-cap disjointness, checks the branch handoff, and regenerates the certificate JSON byte for byte.

Independent audits. A separately organized NumPy/binary64 program reconstructs the geometry numerically. A base-R program checks the archived finite decimals and structural identities without importing Python. Neither audit is used as proof authority. The Python test suite also reconstructs theorem-critical objects, and seven adversarial tests alter one lower-row endpoint, one fan radius, one multiplier, one section-hull vertex, one cap bracket, one disjointness threshold, and one theorem digit. Every altered object is rejected.

Paper reconciliation. The displayed certificate-critical constants and Appendix A are generated from the machine-readable JSON by `paper/reconcile_certificate.py`. A stale or manually divergent TeX include fails verification. The certificate used for this manuscript has SHA-256

7ea121aad900ce83798d7b1eef12d23a89a309622dbafb7ed50ef5fb016006a1.

Trusted components. The finite proof trusts the semantics of CPython integer arithmetic, `fractions.Fraction`, JSON and NPZ parsing, the small exact verifier source, and the operating system's faithful reading of checksum-protected files. It does not trust binary floating point for a theorem inequality. Typesetting software and generated PDF bytes are outside the mathematical trust boundary.

10. SCOPE AND LIMITATIONS

The result is a universal lower-bound certificate. It does not prove that the coefficient is sharp, characterize equality, identify a minimizing body, or establish Meissner extremality. The explicit Meissner volume remains an upper bound for the unknown minimum, not a consequence of this certificate. The paper also makes no claim of peer review or external independent reproduction.

The present result combines a human-readable geometric reduction with a finite exact computation. Future work may seek stronger circumradius-local estimates or alternative certificates, but no such refinement is used in the theorem proved here.

TABLE 2. Principal certificate census.

Object	Count or certified value
Circumradius split	$\frac{305695601}{500000000}$
Lower-branch rows	12
Fan rays	98,306
Fan triangular cones	196,608
Fan stabilizer orbits	24,833
Changed fan vertices	18,229
Minimum exact fan margin	0.0000000002888904207092857876733697822441...
Exact section hulls	6
Antipodal allocation brackets	3
Near-Jung handoff margin	0.0000021074345301574846190550...

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This manuscript and its verification package were developed in an AI-assisted research workflow. The human author is responsible for the claims, source selection, release decisions, and correction of errors. Detailed disclosure and provenance documentation will accompany the public release.

REFERENCES

- [1] W. Blaschke, *Konvexe Bereiche gegebener konstanter Breite und kleinsten Inhalts*, Math. Ann. 76 (1915), 504–513.
- [2] G. D. Chakerian, *Sets of constant width*, Pacific J. Math. 19 (1966), no. 1, 13–21.
- [3] G. D. Chakerian and H. Groemer, *Convex bodies of constant width*, in *Convexity and its Applications*, Birkhäuser, 1983, 49–96.
- [4] E. M. Harrell II, *A direct proof of a theorem of Blaschke and Lebesgue*, J. Geom. Anal. 12 (2002), no. 1, 81–88.
- [5] HYRA, *A certified geometric lower bound for the three-dimensional Blaschke–Lebesgue problem*, manuscript dated August 15, 2026.
- [6] H. Jung, *Ueber die kleinste Kugel, die eine räumliche Figur einschliesst*, J. Reine Angew. Math. 123 (1901), 241–257.
- [7] B. Kawohl and C. Weber, *Meissner’s mysterious bodies*, Math. Intelligencer 33 (2011), no. 3, 94–101.
- [8] T. Lachand–Robert and E. Oudet, *Bodies of constant width in arbitrary dimension*, Math. Nachr. 280 (2007), no. 7, 740–750.
- [9] H. Lebesgue, *Sur le problème des isopérimètres et sur les domaines de largeur constante*, Bull. Soc. Math. France 42 (1914), 72–76.
- [10] H. Martini, L. Montejano, and D. Oliveros, *Bodies of Constant Width: An Introduction to Convex Geometry with Applications*, Birkhäuser/Springer, 2019.
- [11] A. Nishioka, *An improved lower bound for the three-dimensional Blaschke–Lebesgue problem from spectral and dual perspectives*, J. Math. Anal. Appl. 566 (2027), no. 1, article 131038; doi:10.1016/j.jmaa.2026.131038; arXiv:2606.01754.
- [12] R. Schneider, *Convex Bodies: The Brunn–Minkowski Theory*, 2nd expanded ed., Cambridge University Press, 2014.

APPENDIX A. COMPLETE LOWER-CIRCUMRADIUS PARTITION

Table 3: Gap-free lower-circumradius certificate. The final column is the displayed width-normalized volume coefficient; exact coefficients of π are stored in the certificate JSON.

Case	Circumradius interval	Exact mechanism	Volume lower bound
0	$[\frac{1}{2}, \frac{3}{5}]$	analytic tangent bound	0.417357494296972

Continued on next page

Case	Circumradius interval	Exact mechanism	Volume lower bound
01A	$\left[\frac{3}{5}, \frac{6047}{10000}\right]$	exact radial caps	0.416797024215342
01B	$\left[\frac{6047}{10000}, \frac{3047}{5000}\right]$	exact radial caps	0.412421390558881
02A	$\left[\frac{3047}{5000}, \frac{61019}{10000}\right]$	exact radial caps	0.412690926230586
02B	$\left[\frac{61019}{10000}, \frac{30549}{5000}\right]$	exact radial caps	0.411993918231919
03A1R1	$\left[\frac{30549}{50000}, \frac{30553}{50000}\right]$	doubled angular grid	0.412144899831404
03A1R2	$\left[\frac{30553}{50000}, \frac{30557}{50000}\right]$	doubled angular grid	0.412075419676059
03A1R3	$\left[\frac{30557}{50000}, \frac{30561}{50000}\right]$	doubled angular grid	0.412005994020500
03A1R4	$\left[\frac{30561}{50000}, \frac{6113}{10000}\right]$	doubled angular grid	0.411936623442487
03A2	$\left[\frac{6113}{10000}, \frac{244541}{400000}\right]$	exact radial caps	0.411898771358480
03B01	$\left[\frac{244541}{400000}, \frac{76421809}{125000000}\right]$	exact radial caps	0.411890321028654
03B02	$\left[\frac{76421809}{125000000}, \frac{305695601}{500000000}\right]$	exact radial caps	0.411877563780302

The exact row coefficients, angular and radial grids, positive and negative step counts, tangent radii, and the four refined proof arrays are stored under `certificate/certificates/`. Decimal entries in Table 3 are generated displays; the verifier compares exact rational coefficients of π .

APPENDIX B. PAPER-FACING EXACT CONSTANTS

For convenient review, the principal width-normalized quantities are

$$\begin{aligned}
R_* &= \frac{305695601}{500000000} = 0.611391202, \\
E_* &= 0.01939912533733453686615863855265\dots, \\
\det H &\geq 0.99011227429940592416253966614320\dots, \\
C_{\text{fan}} &= 0.5851824745606993360954839072\dots, \\
V_{\text{core}} &\geq 0.4117357103127771735489695298\dots, \\
V_{\text{caps}} &\geq 0.0001439609020552581838284512\dots, \\
V_{\text{near}} &\geq 0.4118796712148324317327979810\dots, \\
V_{\text{universal}} &= \frac{26220940089713\pi}{200000000000000} = 0.4118775637803022742481789259828398710025\dots
\end{aligned}$$

Every displayed value in this list is regenerated from the certificate JSON.

APPENDIX C. MACHINE-READABLE PROOF OBJECTS

Path	Role
<code>universal_contact_bound_certificate.json</code>	Theorem coefficient, exact branch data, fan and cap summaries, and claim boundary.
<code>refined_lower_cell_1.npz</code> <code>refined_lower_cell_4.npz</code>	Exact rational propagation chains replacing the insufficient lower-row cell.
<code>universal_contact_fan.npz</code>	Integer radial data and dyadic multiplier numerators for the adjusted fan.
<code>reference_near_jung_fan.npz</code>	Regression fan used to check nesting and inherited multipliers.
<code>all_axis_pair_section_hulls.tsv</code>	Six exact section polygons used in the cone-cap calculation.
<code>fan_orbit_margin_audit.tsv</code>	One exact margin record for each stabilizer orbit.
<code>audit_constants.csv</code>	Exact/decimal bridge for the independent numerical audits.

The repository checksum manifests authenticate these objects before and after replay. The exact verifier also rebuilds derived TSV, CSV, and JSON objects byte for byte.